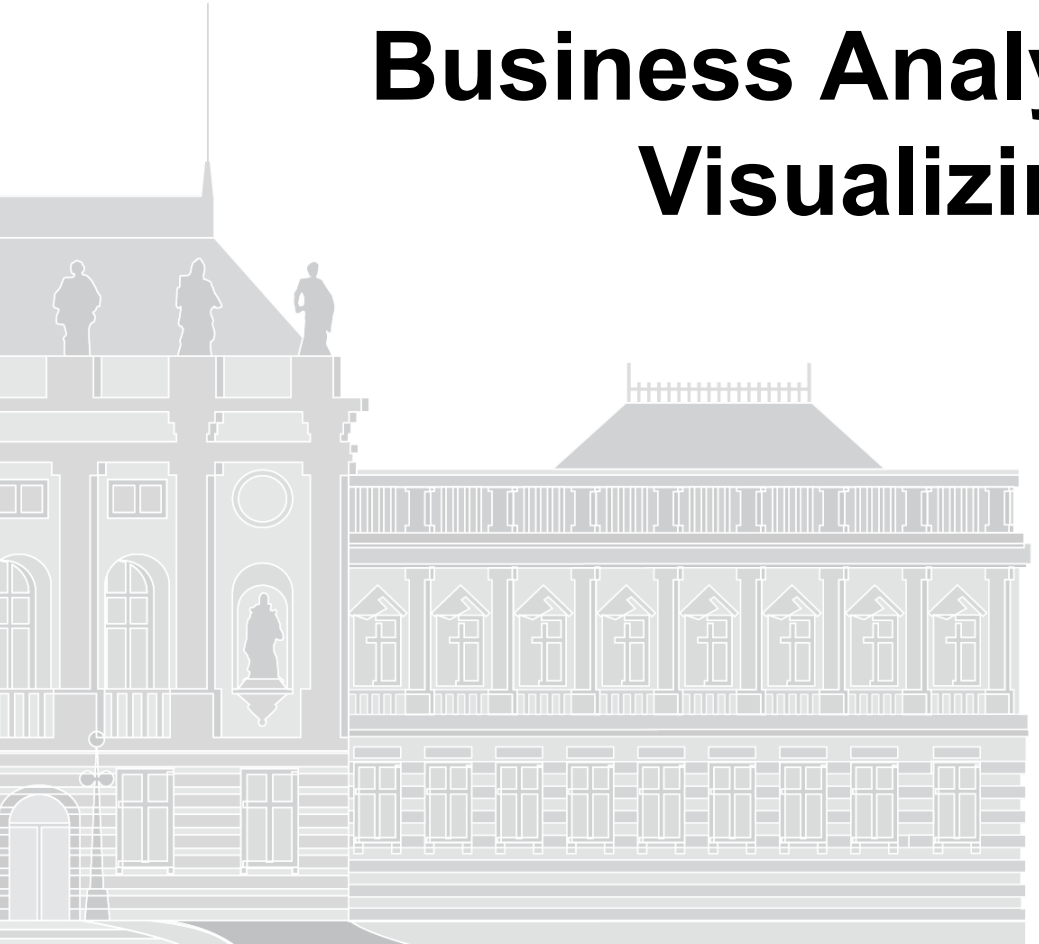
Business Analytics with R – Visualizing Data



Outline

- Why visualize data?
- Goals of the current lecture
- What do we want you to do with Data Visualization? Which insights do we want you to generate?
- Visualizing Data in R

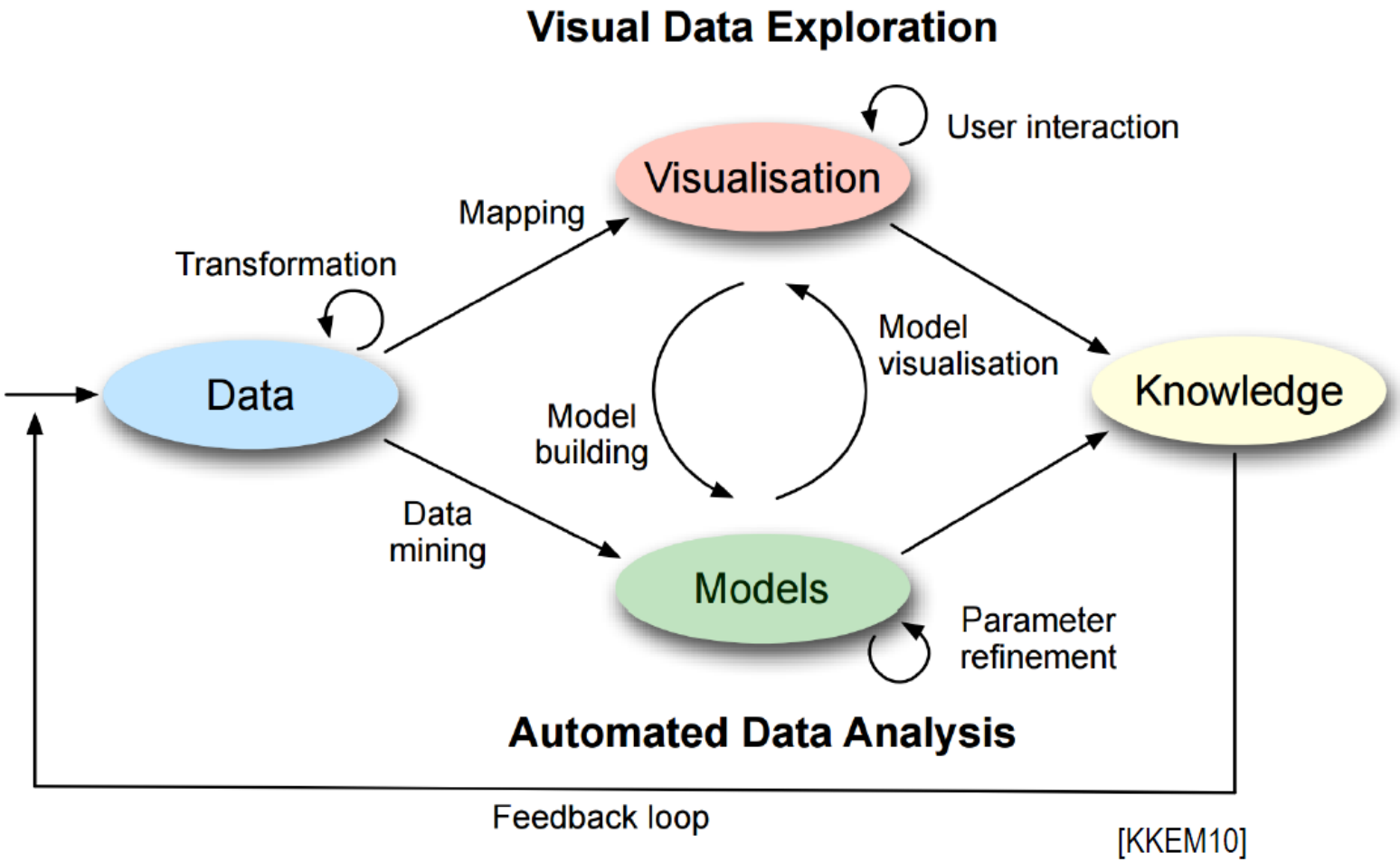
Goals of the current lecture

- Students can gain an overview over data through the use of explorative and graphical techniques.
  - As mentioned earlier, Data Visualization can have many different uses. Each use case has different requirements and needs to be approached differently. This means you may need different tools, depending on what the exact goal of the project you are working on is.
  - The focus of this lecture is on exploring data visually with the goal of using the resulting insights for the development of machine learning models.
  - Note:
    - You will be able to use the tools, technologies and methods in several other contexts as well.

Why visualize data?

- One (or more) of several reasons:
  1. Communicating insights to decision makers (i.e. Management, …)
  2. Data Analysis
     1. Developing or Improving Machine Learning Models
     2. Conducting a Visual Analysis of Data and make deductions from it directly.

Visual Analytics Process

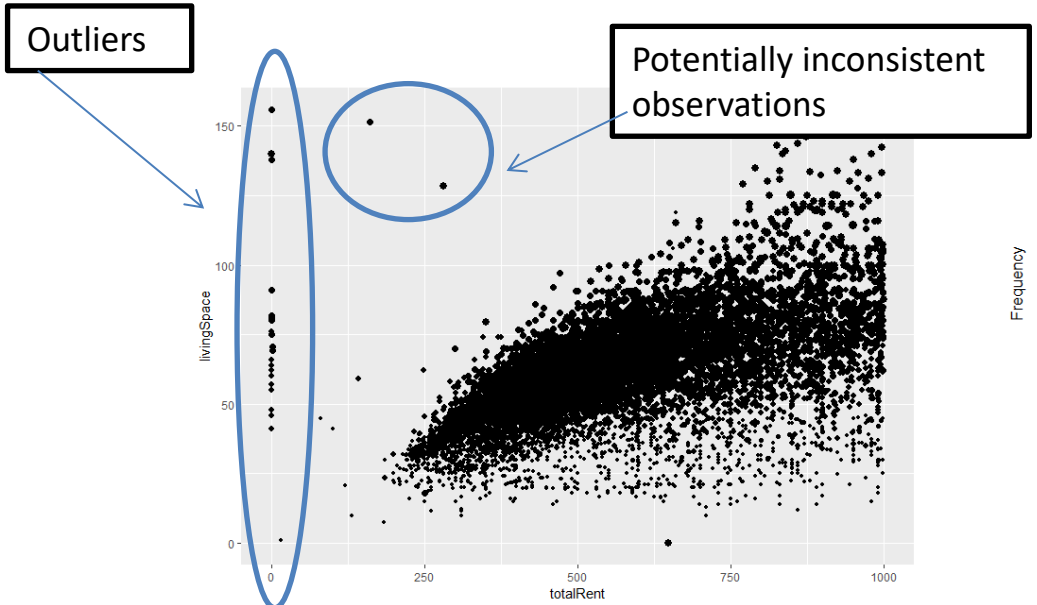


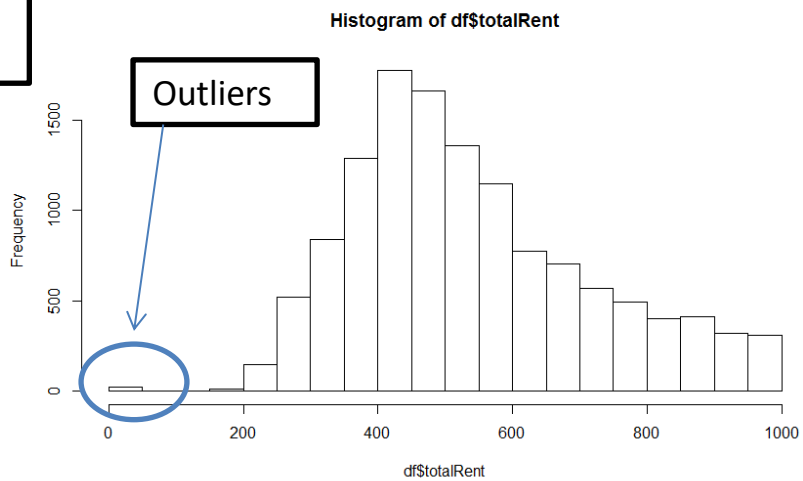
Visual Analytics Workflow by Keim (2010)

What do we want you to do with Data Visualization? Which insights do we want you to generate?

How to use Data Visualization

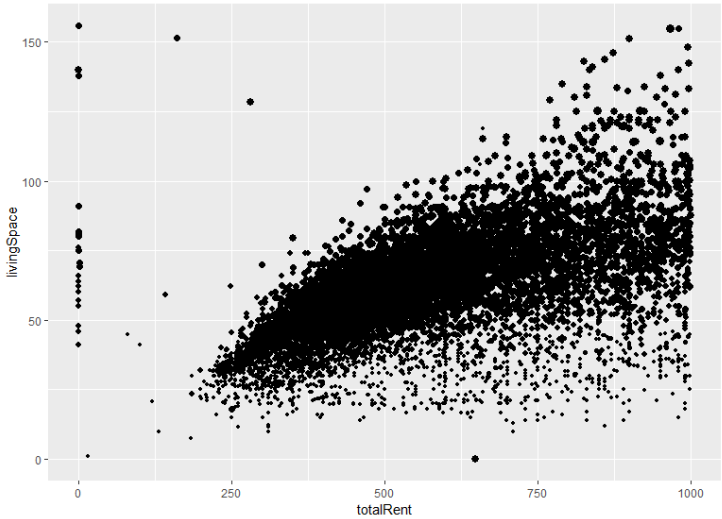
1. “Rear Mirror” – Data Visualization is an easy and effective way to see how well you have cleaned your dataset.
   1. Histograms and scatter plots, for instance, can help you with detecting any remaining outliers.
   2. Scatter plots can give you indications about potential inconsistencies.
2. Looking ahead – Data Visualization can make relationships between variables visible.





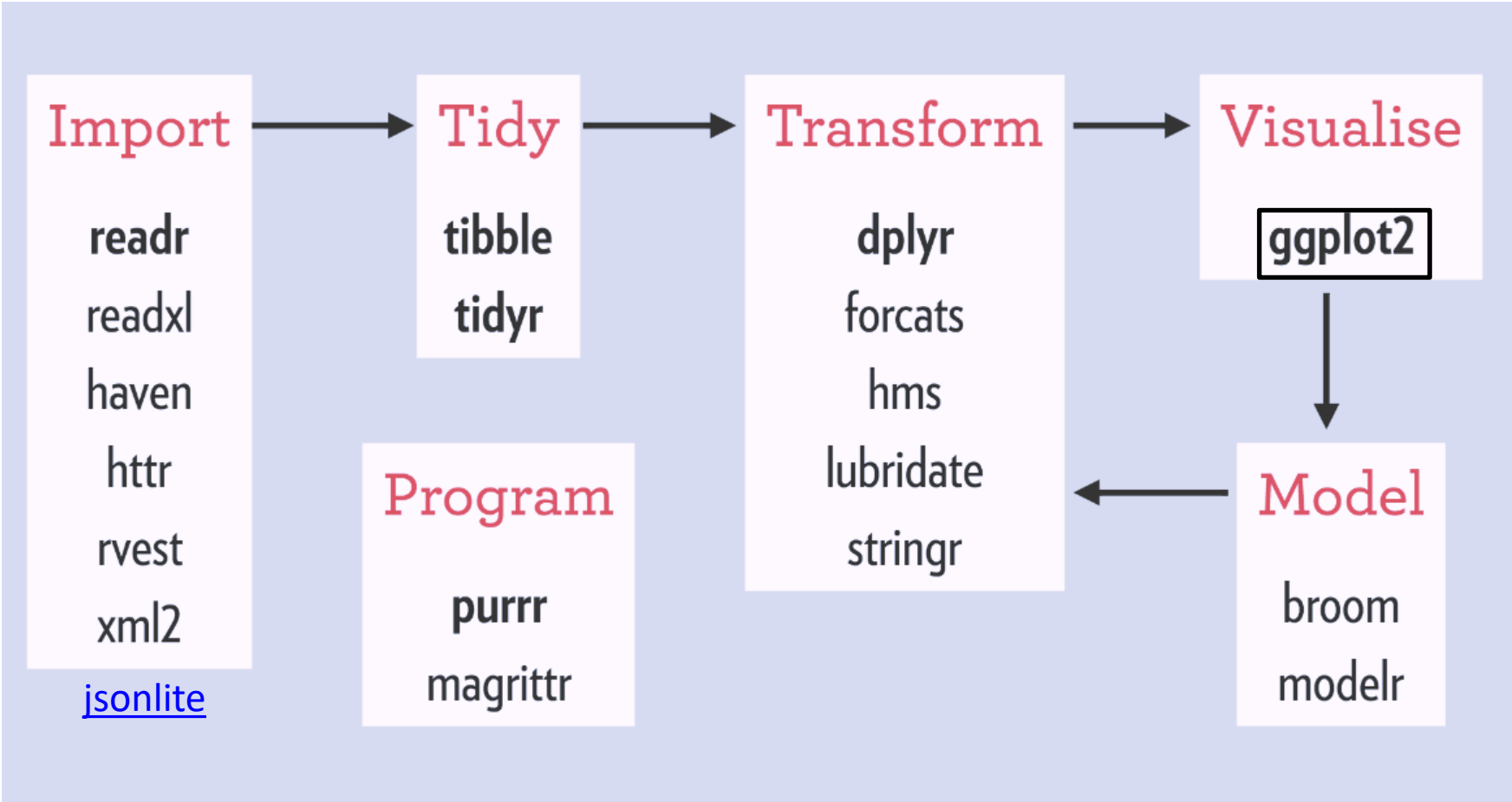
How to use Data Visualization

1. “Rear Mirror” – Data Visualization is an easy and effective way to see how well you have cleaned your dataset.
2. Looking ahead – Data Visualization can make relationships between variables visible.
   1. The scatter plot shows an upward relationship between living space and total rent.



Visualizating Data in R

Tidyverse

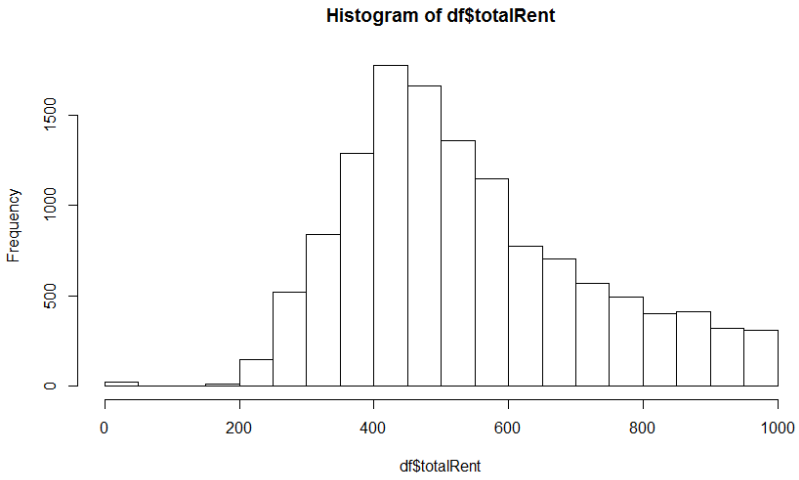

Import
readr
readxl
haven
httr
rvest
xml2
jsonlite
Tidy
tibble
tidyr
Transform
dplyr
forcats
hms
lubridate
stringr
Visualise
ggplot2
Model
broom
modelr
Program
purrr
magrittr

ggplot2 package

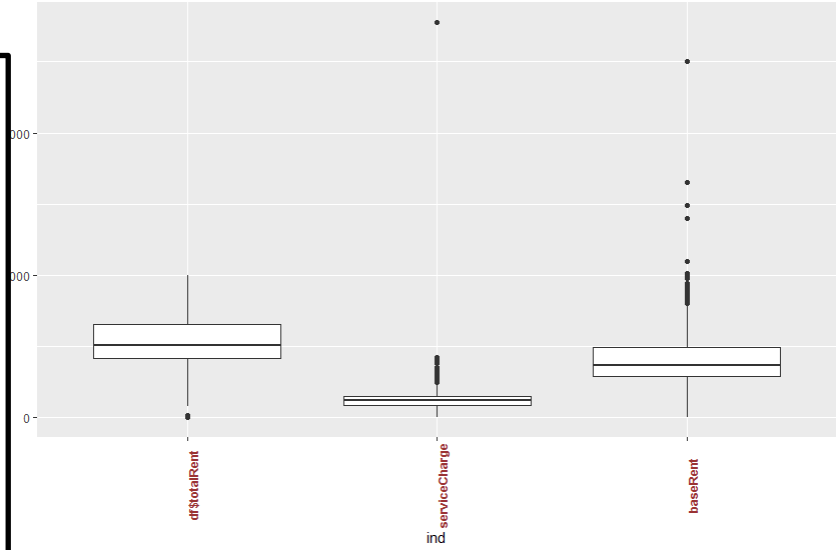
- Ggplot2 is an R package that helps you with exploring datasets visually. It is part of the tidyverse package that we have already seen earlier.
- Cheat Sheet:
  - https://github.com/rstudio/cheatsheets/blob/master/data-visualization-2.1.pdf

Getting an overview

- Distribution Plots
  - Histograms
  - Boxplots

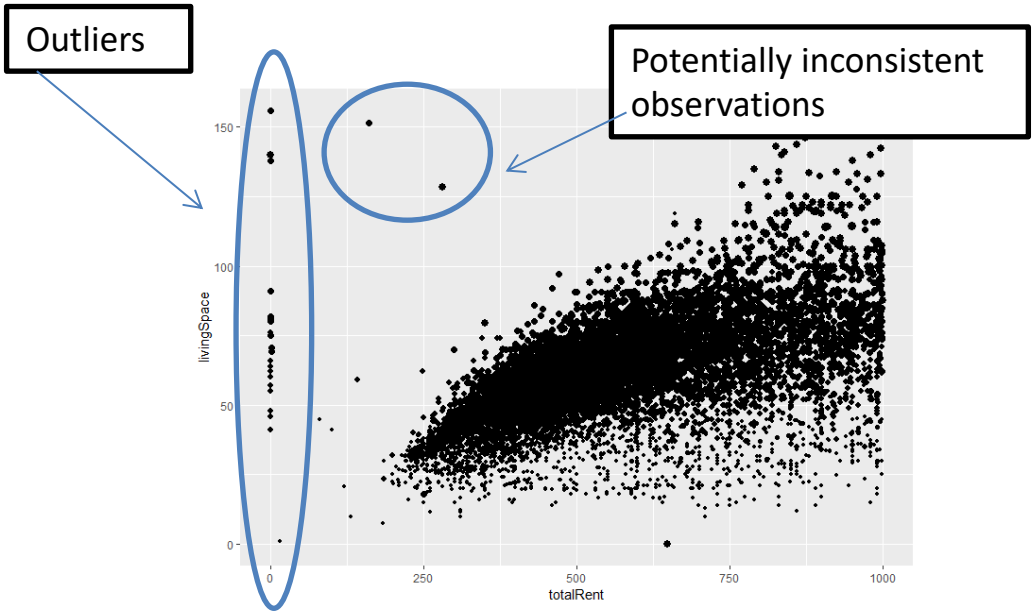


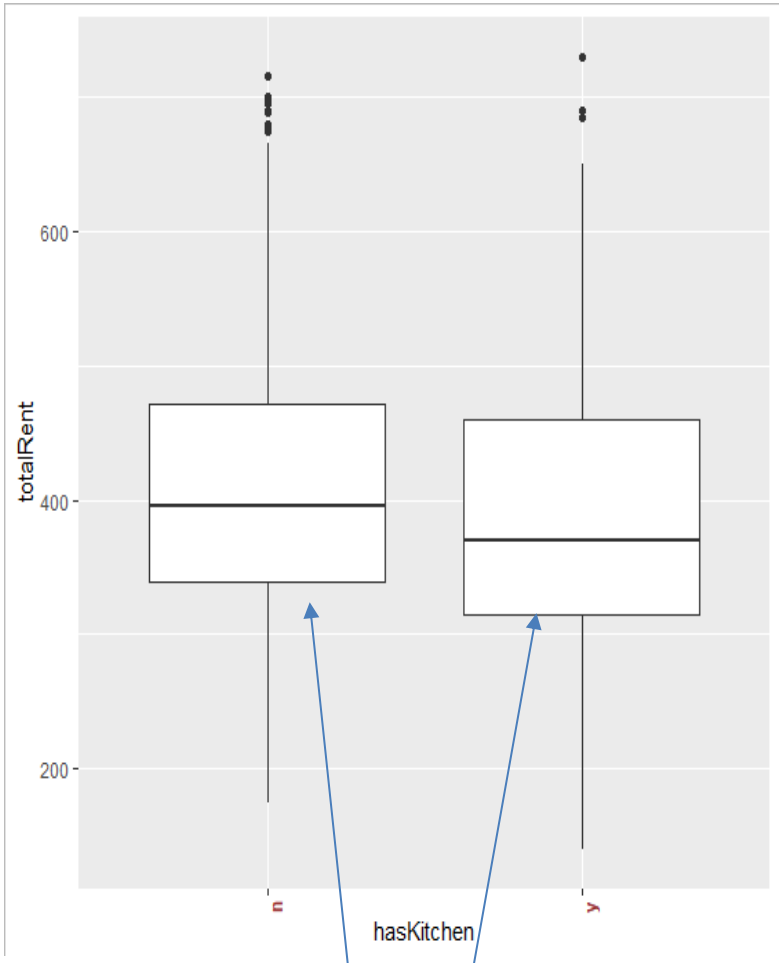

```
# Histogram
> hist(df$totalRent)

# Boxplot
> library(tidyverse)
> df.1 <- as.data.frame(df$totalRent)
> df.1$serviceCharge <- df$serviceCharge
> df.1$baseRent <- df$baseRent
> p <- ggplot(stack(df.1), aes(x = ind, y = values)) +
        geom_boxplot() +
        theme(axis.text.x = element_text(face = "bold",
            color = "#993333", angle = 90))
> p
```



Getting an overview

- Sometimes, these visualizations reveal somewhat counterintuitive relationships or mistakes

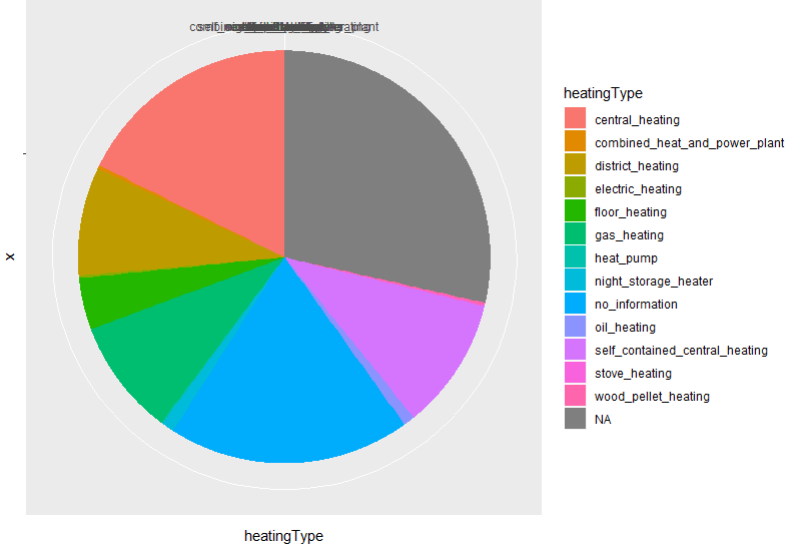




Rent for apartments without a kitchen is higher than for apartments with a kitchen

Getting an overview

- Distribution Plots
  - Histograms
  - Boxplots
- Pie Charts

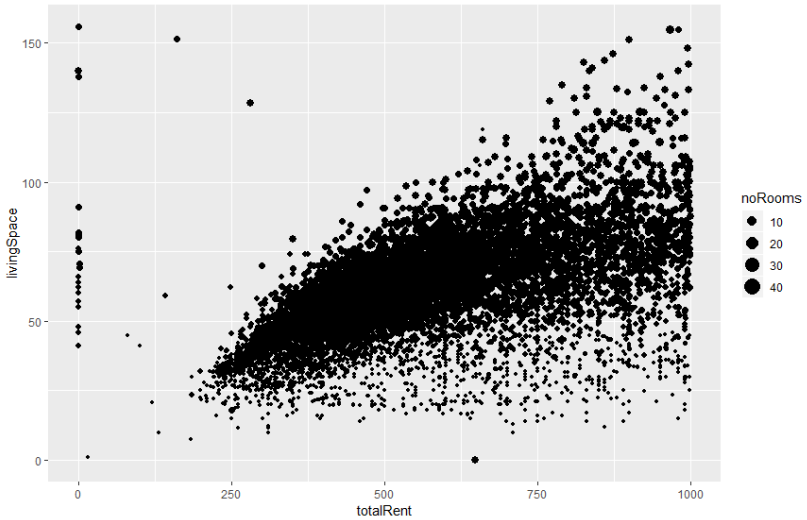


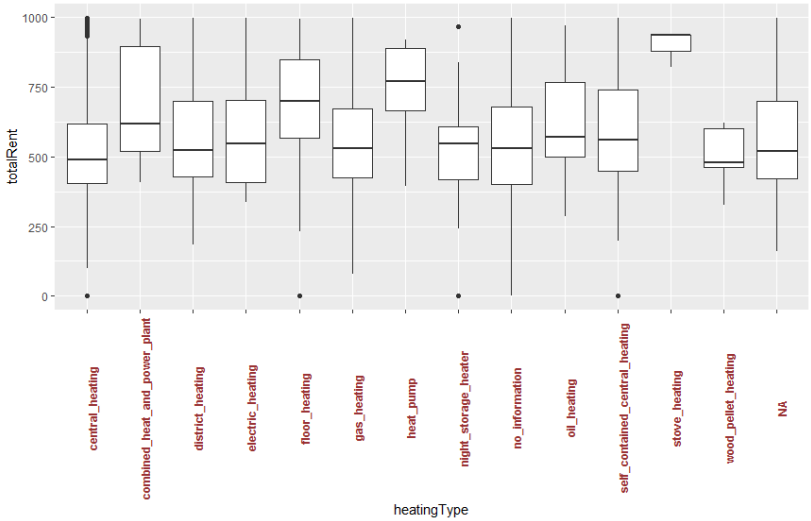

```
# Pie Charts
> library(tidyverse)
bp<- ggplot(df, aes(x="", y=heatingType,fill=heatingType))+
      geom_bar(width = 1, stat = "identity")
# bp
> pie <- bp + coord_polar("y", start=0)
> pie
```

Zooming in – Relationships between variables

- Scatter Plots
- Box Plots

```
> library(tidyverse)

# Scatterplot
> scatter <- ggplot(df, aes(x=totalRent, y=livingSpace)) +
        geom_point(aes(size=noRooms))
> scatter

# Boxplot
> p <- ggplot(df, aes(x=heatingType, y=totalRent)) +
      geom_boxplot() +
      theme(axis.text.x = element_text(face = "bold",
         color = "#993333", angle = 90))
> p
```

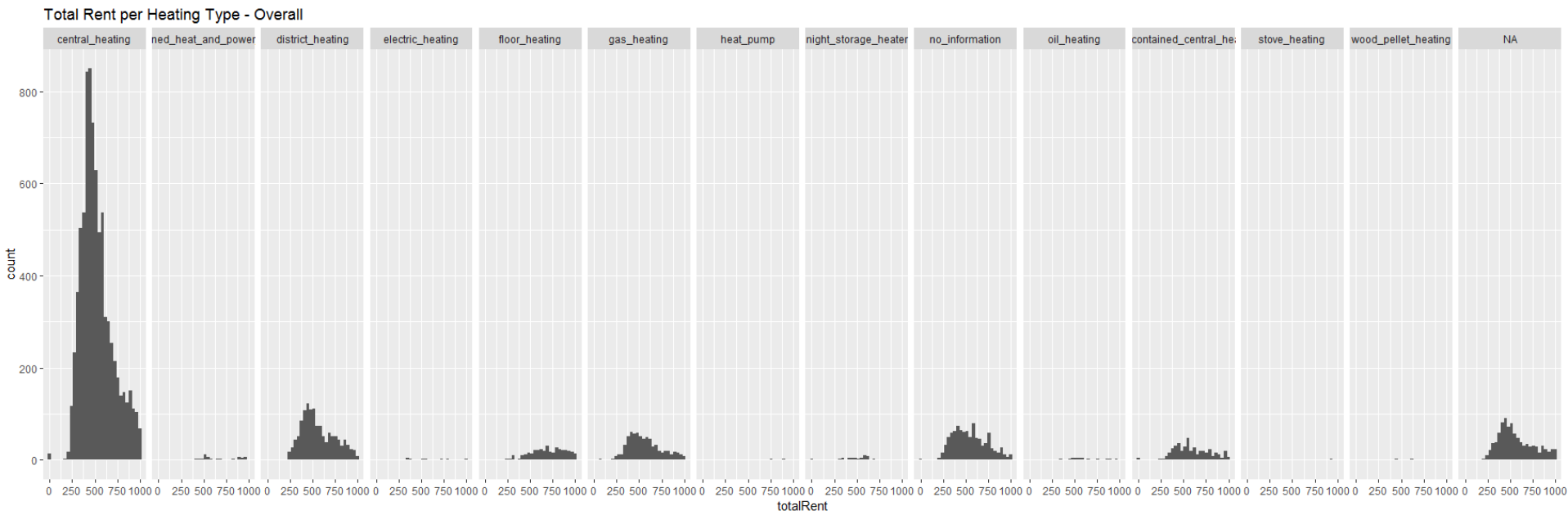




Zooming in – Relationships between variables

- Scatter Plots
- Box Plots
- Bar Charts

```
> library(tidyverse)

# Scatterplot
> k <- ggplot(df, aes(totalRent, stat(count)))+
    geom_histogram(binwidth = 0.2) +
    geom_histogram(aes(y = stat(count))) +
    ggtitle("Total Rent per Heating Type - Overall")
> k + facet_grid(. ~ heatingType)
```



Questions?